

Department of Computer Science & Engineering

B.Tech Semester V Course Syllabus & Lab Manual Guide

Course 1: CS502 - Database Management Systems (4 Credits)

Course Objectives: Understand relational database design, query optimization, indexing, and transaction management.

- **Unit 1 - Relational Model:** Relational algebra, tuple relational calculus, integrity constraints.
- **Unit 2 - SQL and Query Processing:** Complex queries, nested subqueries, views, triggers, and cost-based query optimization.
- **Unit 3 - Normalization:** Functional dependencies, 1NF, 2NF, 3NF, BCNF, lossless join decomposition.
- **Unit 4 - Transactions & Concurrency:** ACID properties, serializability, two-phase locking (2PL), deadlock prevention.
- **Unit 5 - Storage & Indexing:** B-Trees, B+ Trees, hashing techniques.
- **Prescribed Textbook:** *Database System Concepts* by Silberschatz, Korth, and Sudarshan, 7th Edition, McGraw-Hill.

Course 2: CS503 - Operating Systems (4 Credits)

Key Topics:

- Process scheduling (FCFS, SJF, Round Robin, Multilevel Feedback Queue).
- Synchronization: Critical section problem, Peterson algorithm, semaphores, mutexes, dining philosophers problem.
- Deadlocks: Resource allocation graph, Banker's algorithm for deadlock avoidance.
- Memory Management: Paging, segmentation, page replacement algorithms (FIFO, LRU, Optimal).

Course 3: CS508 - Database & OS Practical Laboratory Rules

- **Lab Attendance:** Minimum 80% practical attendance is strictly enforced.
- **Lab Records:** Completed lab notebooks with experiment results must be submitted every Monday by 10:00 AM.
- **Evaluation Scheme:** Continuous assessment (40 marks), Internal Viva (20 marks), End-semester practical exam (40 marks).